

Windows DFIR Incident Timeline Investigation

Northstar Ridge Financial Services

MULTI-SOURCE EVIDENCE • TIMELINE RECONSTRUCTION • INCIDENT RESPONSE

Organization	Northstar Ridge Financial Services (fictional)
Case Type	Windows DFIR / Incident Timeline Reconstruction
Analyst	Ryan Bryant
Primary Host	NFG-FIN-WS07
Secondary Host	NFG-FIN-APP01 (requires further investigation)
Primary Identity	finance.user
Evidence Window	September 17, 2026
Classification	Portfolio demonstration - no production systems

Safety and Evidence Note

This report documents a fictional Windows incident-response exercise. All hostnames, identities, IP addresses, commands, hashes, and events are synthetic or reserved for documentation/testing. The analysis distinguishes confirmed evidence from likely conclusions and unresolved hypotheses.

1. Executive Summary

The investigation identified a sequence strongly consistent with unauthorized remote access to NFG-FIN-WS07 using the finance.user identity. Three failed remote interactive logons from 198.51.100.24 were followed by a successful remote interactive logon from the same source. Activity immediately afterward included host and account discovery, PowerShell execution from a temporary directory, internal network activity, creation of new files, service-based persistence, and suspicious outbound communications.

The evidence confirms compromise-related activity on NFG-FIN-WS07 and persistence through the SystemHelper Windows service running as LocalSystem. Activity toward NFG-FIN-APP01 supports suspected attempted lateral movement; however, successful compromise of that server has not been established.

The available evidence does not determine how the finance.user credentials were obtained or how the privileges necessary to install a LocalSystem service were acquired. Those questions require additional endpoint, PowerShell, authentication, process-tree, and network evidence.

Assessment Item	Current Conclusion	Confidence
Unauthorized remote access	Likely compromise of finance.user used for remote access	High
Discovery activity	Account, administrator-group, network, and service discovery confirmed	High
Suspicious execution	PowerShell script executed from user Temp with ExecutionPolicy Bypass	High
Lateral movement	Attempted/suspected movement toward NFG-FIN-APP01; successful compromise not established	Moderate-High
Persistence	SystemHelper service installed, started automatically, and ran as LocalSystem	High
Privilege escalation mechanism	Context transitioned to SYSTEM, but method is unknown	Moderate
Command-and-control	Outbound communications are suspicious; confirmed C2 not established	Moderate

2. Incident Scope

Category	Assessment
Confirmed involved host	NFG-FIN-WS07
Host requiring investigation	NFG-FIN-APP01
Confirmed affected identity	finance.user
Observed system context	SYSTEM / LocalSystem
Observed legitimate identities	svc-finance and it.admin; no current evidence of compromise
External authentication source	198.51.100.24
Suspicious external destinations	203.0.113.77 and 203.0.113.88
Confirmed persistence	SystemHelper service / svc-helper.exe
Suspected lateral movement	NFG-FIN-WS07 toward NFG-FIN-APP01
Confirmed secondary-host compromise	Not established

2.1. Evidence Sources

- Windows authentication events (4624, 4625, 4634).
- Windows process-creation telemetry (4688).
- File-system activity with synthetic SHA-256 hashes.
- Network connection telemetry, including allowed and blocked flows.
- Windows service installation and state-change events (7045, 7036).
- Microsoft Defender-style security alerts.

Analyst Standard

A security alert is treated as an investigative lead, not proof. Findings are based on correlation across raw authentication, process, file, network, service, and security-alert evidence.

3. Incident Timeline

The following chronology focuses on the suspicious activity window from the first failed remote logon through session termination and the initial administrator response. Times are UTC.

Time	Evidence Source	Host / User	Raw Activity
13:41:58	Authentication	NFG-FIN-WS07 / finance.user	Failed remote interactive logon from 198.51.100.24
13:42:09	Authentication	NFG-FIN-WS07 / finance.user	Second failed remote interactive logon from 198.51.100.24
13:42:27	Authentication	NFG-FIN-WS07 / finance.user	Third failed remote interactive logon from 198.51.100.24
13:43:06	Authentication	NFG-FIN-WS07 / finance.user	Successful remote interactive logon from 198.51.100.24
13:43:47	File	NFG-FIN-WS07 / finance.user	update.ps1 created in user AppData\Local\Temp
13:43:55	Process	NFG-FIN-WS07 / finance.user	cmd.exe /c whoami
13:44:08	File	NFG-FIN-WS07 / finance.user	update.ps1 modified shortly before execution
13:44:12	Process	NFG-FIN-WS07 / finance.user	PowerShell executes update.ps1 with -NoProfile and -ExecutionPolicy Bypass
13:44:15	Defender	NFG-FIN-WS07 / finance.user	Suspicious PowerShell execution alert
13:44:39	Process	NFG-FIN-WS07 / finance.user	cmd.exe /c net user
13:45:03	Process	NFG-FIN-WS07 / finance.user	cmd.exe /c net localgroup administrators
13:46:33	Authentication	NFG-FIN-APP01 / finance.user	Network logon from NFG-FIN-WS07 (10.10.20.17)
13:47:18	Process	NFG-FIN-WS07 / finance.user	PowerShell Get-NetTCPConnection
13:48:22	File	NFG-FIN-WS07 / finance.user	C:\Users\Public\system-cache.dat created
13:48:31	Network	NFG-FIN-WS07 / finance.user	PowerShell outbound HTTPS to 203.0.113.77:443
13:48:36	Defender	NFG-FIN-WS07 / finance.user	Unusual outbound PowerShell connection alert
13:50:12	Network	NFG-FIN-WS07 / finance.user	SMB connection attempt to 10.10.20.40:445 blocked
13:51:46	Process	NFG-FIN-WS07 / finance.user	cmd.exe /c sc query
13:52:11	File	NFG-FIN-WS07 / SYSTEM	C:\ProgramData\SystemHelper\svc-helper.exe created
13:52:38	Service	NFG-FIN-WS07 / SYSTEM	SystemHelper installed as automatic LocalSystem service
13:52:44	Defender	NFG-FIN-WS07 / SYSTEM	Unusual service installation alert
13:52:51	Service	NFG-FIN-WS07 / SYSTEM	SystemHelper entered running state
13:53:04	Network	NFG-FIN-WS07 / SYSTEM	svc-helper.exe outbound TCP to 203.0.113.88:8443
13:53:09	Defender	NFG-FIN-WS07 / SYSTEM	Suspicious service network activity alert
13:58:33	File	NFG-FIN-WS07 / finance.user	update.ps1 deleted
14:03:18	Authentication	NFG-FIN-WS07 / finance.user	Remote session from 198.51.100.24 ended
14:15:02	Authentication	NFG-FIN-WS07 / it.admin	Approved administrator remote session
14:19:26	Service	NFG-FIN-WS07 / it.admin	SystemHelper stopped during investigation
14:20:11	Defender	NFG-FIN-WS07 / it.admin	SystemHelper stop recorded as response action

4. Investigative Findings

4.1. Unauthorized Remote Authentication

Confidence: High

Evidence: Three failed remote interactive logons from 198.51.100.24 were followed by a successful remote interactive logon from the same source using finance.user.

Assessment: The sequence is strongly consistent with unauthorized use of valid credentials. The evidence does not establish how those credentials were obtained.

4.2. Post-Access Discovery

Confidence: High

Evidence: Commands included whoami, net user, net localgroup administrators, Get-NetTCPConnection, and sc query.

Assessment: The activity confirms user/host discovery, account discovery, privileged-group enumeration, network discovery, and service discovery. The local Administrators group was enumerated, not modified.

4.3. Suspicious PowerShell Execution

Confidence: High

Evidence: PowerShell executed C:\Users\finance.user\AppData\Local\Temp\update.ps1 using -NoProfile and -ExecutionPolicy Bypass. The script was created/modified immediately before execution and deleted later.

Assessment: The sequence is suspicious and temporally linked to the remote session. The script's complete purpose cannot be established without script contents, Script Block Logging, or transcript evidence.

4.4. Attempted Lateral Movement

Confidence: Moderate-High

Evidence: A network logon from NFG-FIN-WS07 reached NFG-FIN-APP01, and a later SMB attempt toward the Finance application environment was blocked.

Assessment: Attempted or suspected lateral movement is supported. Successful execution or compromise of NFG-FIN-APP01 has not been established.

4.5. Service-Based Persistence

Confidence: High

Evidence: svc-helper.exe was created under ProgramData, installed as the automatic SystemHelper service, started under LocalSystem, and then initiated outbound network activity.

Assessment: Persistence behavior is confirmed through Windows service installation. The service should be preserved for forensic analysis before removal.

4.6. Privilege Context Transition

Confidence: Moderate

Evidence: Earlier activity occurred under finance.user. Later file/service activity occurred as SYSTEM / LocalSystem.

Assessment: A privilege-context transition is confirmed, but the mechanism is not. The evidence does not prove a specific privilege-escalation technique.

4.7. Suspicious Outbound Communications

Confidence: Moderate-High

Evidence: PowerShell contacted 203.0.113.77 over HTTPS, and the newly installed service contacted 203.0.113.88 over TCP/8443.

Assessment: The behavior is consistent with possible command-and-control style activity, but the available evidence does not confirm destination ownership, traffic contents, or C2 protocol.

5. Evidence Confidence Matrix

Classification	Supported Statements
Confirmed	Repeated failed remote logons; successful remote session; discovery commands; PowerShell script execution; file creation; blocked SMB attempt; new service installation; LocalSystem execution; outbound network communications; temporary script deletion.
Likely	finance.user credentials were compromised; the remote session was unauthorized; the actor attempted lateral movement; SystemHelper was intended to preserve access or execution.
Unresolved / Hypothesis	Phishing as the credential-theft method; successful privilege-escalation technique; successful compromise of NFG-FIN-APP01; confirmed command-and-control infrastructure; malware-family attribution; data exfiltration.
Key Forensic Gap The most important unresolved window is 13:51:46-13:52:11 UTC, where activity transitions from finance.user service discovery to creation of svc-helper.exe under SYSTEM. Additional process-tree, PowerShell, privileged-logon, Sysmon/EDR, and service-control evidence is required to explain this transition.	

6. Immediate Containment and Response Priorities

Priority	Action	Rationale
1	Isolate NFG-FIN-WS07 from normal network access while preserving approved IR/management connectivity.	Stops further external or lateral communication while preserving evidence access.
2	Contain finance.user; revoke active sessions/tokens where applicable; reset credentials and verify MFA.	Removes the identity used during the suspicious remote session.
3	Preserve evidence for SystemHelper and svc-helper.exe before deletion; collect hashes, service configuration, process data, memory if available, and related logs.	Maintains evidence needed to explain persistence and privilege context.
4	Block or closely monitor suspicious external destinations 203.0.113.77 and 203.0.113.88 at appropriate network controls.	Reduces continued outbound communication and supports enterprise-wide hunting.
5	Investigate NFG-FIN-APP01 for authentication, process, file, SMB, service, and EDR evidence before declaring it compromised or isolating it.	Avoids overstating secondary-host compromise while addressing the lateral-movement possibility.
6	Expand retrospective review to at least the previous 24 hours across authentication, PowerShell, process, network, Defender/EDR, and service telemetry.	Helps identify earlier access, credential theft, or related systems.

6.1. Additional Evidence Requested

- PowerShell Script Block Logging and PowerShell transcripts.
- Windows privileged-logon and special-privilege events.
- EDR/Defender process trees showing parent/child relationships.
- Sysmon telemetry, if available, for process, file, network, and registry activity.
- Service Control Manager activity and the process responsible for creating SystemHelper.
- Packet capture, firewall, proxy, DNS, or flow logs for 13:41-13:54 UTC.
- Browser history and email records to test the phishing/credential-theft hypothesis.
- Entra ID sign-in and MFA telemetry if cloud authentication is relevant to the account.
- Registry, scheduled-task, startup-folder, and other persistence artifacts.
- Memory capture and credential-access evidence from the affected workstation.

7. MITRE ATT&CK Context

The following mappings describe behaviors observed or reasonably supported by the synthetic evidence. They are contextual mappings, not attribution or proof of a specific threat actor.

Observed Behavior	Technique	ATT&CK ID
Remote interactive access	Remote Services	T1021
PowerShell execution	Command and Scripting Interpreter: PowerShell	T1059.001
whoami	System Owner/User Discovery	T1033
net user	Account Discovery	T1087
net localgroup administrators	Permission Groups Discovery: Local Groups	T1069.001
Get-NetTCPConnection	System Network Connections Discovery	T1049
sc query	System Service Discovery	T1007
SystemHelper service	Create or Modify System Process: Windows Service	T1543.003
Attempted SMB access toward Finance server	Remote Services: SMB/Windows Admin Shares (contextual)	T1021.002

8. Final Analyst Conclusion

Conclusion

The investigation identified a sequence strongly consistent with unauthorized access to NFG-FIN-WS07 using the finance.user identity. Failed remote authentication attempts from an external source were followed by a successful remote session and subsequent discovery, PowerShell execution, internal network activity, file creation, and service-based persistence.

A newly created executable was installed as the automatic SystemHelper service and executed under LocalSystem before establishing an outbound connection. Activity toward the Finance application environment supports suspected attempted lateral movement; however, successful compromise of NFG-FIN-APP01 has not been established.

The evidence does not currently determine how the user's credentials were obtained or how the privileges required for LocalSystem service installation were acquired. Additional endpoint, authentication, PowerShell, network, and process-tree evidence is required to resolve those questions. Immediate containment should prioritize the affected workstation, compromised identity, persistence mechanism, and suspicious outbound destinations while preserving forensic evidence for continued investigation.

8.1. Skills Demonstrated

- DFIR evidence review and multi-source timeline reconstruction.
- Windows authentication and process-event interpretation.
- PowerShell and service-persistence analysis.
- Network-flow and lateral-movement assessment.
- Evidence-confidence language: confirmed vs. likely vs. unresolved.
- Containment prioritization and forensic-evidence preservation.
- MITRE ATT&CK contextual mapping.
- Stakeholder-ready incident documentation.

8.2. Portfolio Positioning

This report should be presented as a fictional DFIR investigation exercise. It demonstrates analyst reasoning, evidence correlation, confidence discipline, and incident-response decision making rather than a claim of investigating a real customer or production environment.